

SUNCLAVE · REQUIREMENTS

SunClave requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Apply SCL-DDR-001 (R1, R13 and R17 redefined, design case completed) and replace the status column with the TRL 3 results of SCL-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with design case and status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

SunClave requirements

These requirements are checked by calculation in SCL-CAL-001 at TRL 3. Targets are still proposals, not yet validated with users, and will be revised after co-design sessions (see SCL-PRB-001). R1, R13 and R17 were restated on 2026-09-25 to apply Amish's decisions in SCL-DDR-001 (altitude option B, and the budget cuts and \$450 budget). The status column gives the TRL 3 result; every value is a paper estimate.

The **design case** is a clear day with 700 W/m² direct normal irradiance (DNI) and the sun 60° above the horizon, 25 °C ambient, wind 2 m/s (below 3 m/s), a site at sea level, and a load of 2.0 kg of unwrapped or single-wrapped solid stainless instruments in a 0.5 kg basket, with 1.5 L of water in a 12 L aluminum cooker.

Safety: These requirements describe a pressure vessel heated by concentrated sunlight, in a research and educational prototype that is not a medical device. R8 to R11 are safety requirements and take precedence over throughput and cost. A PASS against R1 and R12 records conditions; it does not prove sterility.

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status (SCL-CAL-001)
R1	Sterilizing condition (redefined 2026-09-25, SCL-DDR-001 item 1)	Saturated steam in the load zone for an exposure equal to 20 min (unwrapped) or 30 min (wrapped) at 121 °C, using $z = 10$ °C for any lower temperature; never below 115 °C nor above 124 °C in normal operation; the real temperature and hold are recorded and a cycle below 121 °C is never labeled 121 °C	Calculation; later instrumented cycles with biological indicators	Not verifiable at TRL 3. Met by calculation to 2,400 m (120.95 °C and 30.3 min at sea level; 117.7 °C and 63 min at 1,800 m), but the equivalence needs biological-indicator evidence
R2	Load capacity	Basket at least 250 mm diameter by 150 mm deep; 2.0 kg of solid instruments per cycle (about two delivery kits or four suture kits)	Model check	Met: 250 x 150 mm basket in a 12.3 L cooker, 16 mm above the water
R3	Load type	Solid instruments only, unwrapped or single-wrapped in porous wrap; no lumened, hollow or textile loads	Instructions and design review	Met by definition (decided, SCL-DDR-001 item 9)
R4	Cycle time	Cold start to end of hold in 90 min or less in the design case	Energy calculation; later timed cycles	At risk: about 80 min central, 71 to 109 min across scenarios
R5	Daily throughput	3 or more complete cycles on a clear day (DNI 600 W/m ² or more from 09:00 to 15:00 solar time)	Energy calculation	Met: 4 cycles in each case checked; zero on overcast days
R6	Cycle record	Load-zone temperature and chamber pressure logged at 1 s intervals from start to end; cycle ID, date and time, hold duration, minimum hold temperature, equivalent exposure and a pass or fail flag stored for 1,000 or more cycles and shown to the operator at cycle end	Design review; later bench test	Met by design: about 173 kB per cycle, about 23,000 cycles on 4 GB
R7	Measurement accuracy	Temperature within 0.5 °C at 121 °C; pressure within 5 kPa; field check against boiling water at each site	Component datasheets; calibration method at TRL 3	Met on datasheets: 0.44 °C with a 0.05 % reference resistor; 5 kPa. The field check needs a reference barometer
R8	Pressure safety	Working pressure 103.4 kPa (15 psi) gauge set by the cooker's regulator; an independent relief valve set at 125 kPa gauge or less; the cooker's own overpressure plug retained; vessel rated by its maker for the working pressure	Design review; relief capacity calculation	Met: relief 5.1 times and vent 2.6 times the worst steam generation; vessel rating to confirm for the chosen model
R9	Heat input control	Operator can stop heat input within 10 s by turning the dish off the sun, and the dish is shaded when parked	Design review	Met: below 5 % of power at 15° off the sun; parking cover in the BOM
R10	Boil-dry protection			

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status (SCL-CAL-001)
		At least 0.5 L of water left at the end of a cycle in the design case; logger alarm if base temperature exceeds 140 °C	Water budget; later test	At risk: 1.26 L left in the design case, but 0.14 L after a 63 min hold at 1,800 m in strong sun without trimming
R11	Burn and glare protection	No exposed surface above 60 °C within reach during normal operation other than the lid and handles; focal zone guarded; eye protection supplied	Design review	Not met: guard not defined; black base and wall band within reach. Goggles and parking cover added. Options in SCL-DDR-001 item 15
R12	Air removal check	Logger flags any hold where measured temperature is more than 2 °C below the saturation temperature for the measured pressure	Calculation; later test with deliberate air leak	At risk: check uncertainty 0.89 °C; sure to flag about 9 % air
R13	Off-grid logger power (redefined 2026-09-25, SCL-DDR-001 item 2)	Logger runs 3 or more days on its own USB power bank without recharging; the bank recharges from any USB source and does not switch off at the logger's low current	Power budget	Met: about 8.7 days
R14	Tracking effort	Retarget no more often than every 15 min; aim by a shadow gnomon without looking at the sun or the focus	Optics calculation	Not met: power stays within 10 % of on-target for about 13 min. Options in SCL-DDR-001 item 13
R15	Stability	Does not tip in a 10 m/s wind with the dish at any tilt and the cooker loaded	Moment calculation	At risk: tipping factor 1.30 at 15° elevation; rolls on two locked castors
R16	Portability and build	Total mass 40 kg or less; separates into pieces of 20 kg or less; built with hand tools, a drill and bolted joints	Mass estimate; model check	Not met: about 41.0 kg empty; largest piece 19.4 kg (met). Options in SCL-DDR-001 item 14
R17	Cost (redefined 2026-09-25, SCL-DDR-001 item 2)	Parts for one prototype \$450 or less, validation consumables excluded	Priced BOM (bom/bom.csv)	Met: \$430

■ Requirements not met or at risk

- **R11 burn and glare (not met).** No focal-zone guard, and the black lower wall is hot and within reach.
- **R14 tracking (not met).** About 13 min between retargets, not 15 min.
- **R16 mass (not met).** About 41.0 kg against 40 kg with the decided timber stand.
- **R4 cycle time (at risk).** Met in the central case (80 min), missed in the unfavourable one (109 min).
- **R10 boil-dry (at risk).** Long option B holds at altitude in strong sun need the dish trimmed.
- **R12 air-removal check (at risk).** The pressure transducer's span limits the check.
- **R15 stability (at risk).** Thin tipping margin at low sun, and the stand can roll.
- **R1 sterilizing condition (not verifiable at TRL 3).** The equivalent-exposure method needs biological indicators.

■ Assumptions

- Dish aperture 1.4 m (1.539 m²), focal length 500 mm, reflectance 0.85, slope error 10 mrad, absorber absorptance 0.92; optics by ray trace in SCL-CAL-001.
- Saturated steam temperatures from IAPWS-IF97; ICAO standard atmosphere for ambient pressure with altitude.
- Hold times follow CDC Table 7 and the IARC manual cited in SCL-PRB-001. Equivalent hold at lower temperature uses a z-value of 10 °C, which is a common engineering assumption and not a validated claim.
- "Pass" in R6 means the recorded cycle met R1 and R12. It does not prove sterility; biological and chemical indicators remain the reference.